

# SpineSense

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# Project Overview

## The Problem

- Roughly 50–70% of desk workers experience back pain due to poor posture
- Average screen time of 6 hour

## Our Solution

A real time computer vision system that detects and scores posture using pose estimation, a trained fault classifier, and geometric spine analysis, deployed on a Jetson Orin Nano.

# Methodology

## System Pipeline



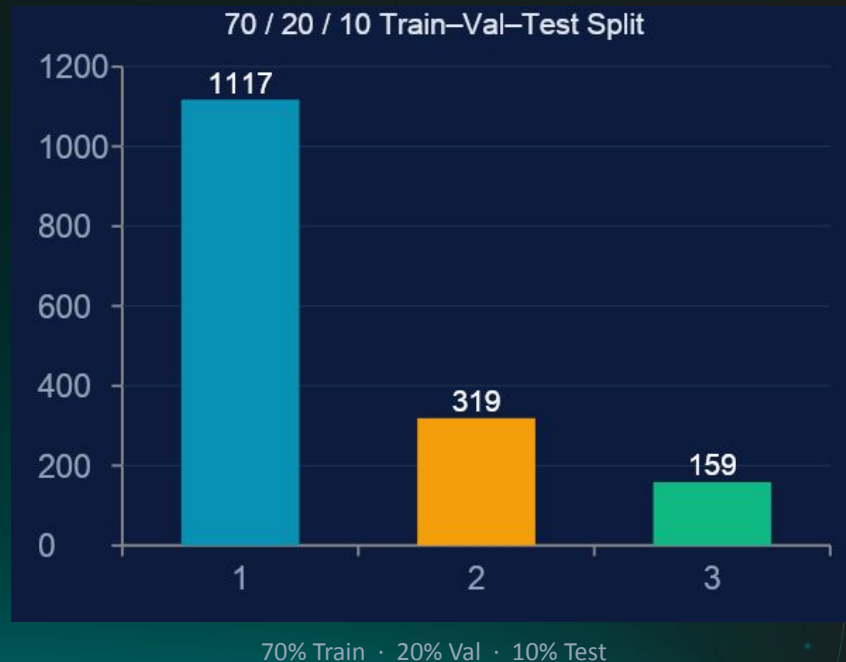
The camera captured images through YOLO 11, then properly labeled the features extracted, into MLP to classify good vs bad and finally display the score.

# Data & Training

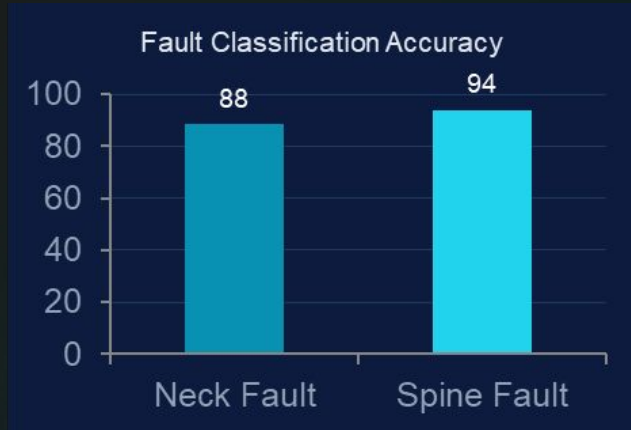
Compiled ~500 images across team members alongside ROBOFLOW dataset with ~1000 images.

Features:

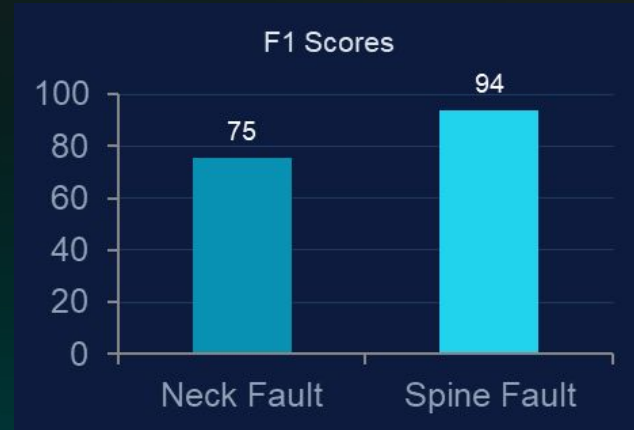
- Neck Angle
- Shoulder
- Head tilt
- Spine Curve



# Metrics



Neck and spine fault exceed 88% accuracy on test data



High F1 scores indicate strong precision and recall balance

# Future Plans

**01**

Mobile & Web App

**02**

Haptic Sensor  
Feedback

**03**

Sitting vs Standing  
Detection

**04**

Multi-Angle  
Camera Support

# Live Demo